

Lab Task 5: N-Grams and Text Generation

Aim:

To develop N-gram models for a word document and perform text generation using NLTK and spaCy in Python.

CODE:

```
# Install (run once)

# !pip install nltk spacy

import nltk

import random

import spacy

from nltk.tokenize import word_tokenize

from nltk.util import ngrams

from collections import defaultdict

# Download required data

nltk.download('punkt')

nltk.download('punkt_tab')

# Load spaCy model

# python -m spacy download en_core_web_sm

nlp = spacy.load("en_core_web_sm")

# Sample text

text = """Natural language processing is a field of artificial intelligence.

It enables computers to understand human language."""

# ♣ Tokenization

tokens = word_tokenize(text.lower())

# ♣ Generate Bigrams

n = 2

bigrams = list(ngrams(tokens, n))

print("Generated Bigrams:")

print(bigrams)

# ♣ Build Model
```

```

model = defaultdict(list)
for w1, w2 in bigrams:
    model[w1].append(w2)
# ♠ Text Generation
def generate_text(seed, num_words):
    current_word = seed.lower()
    output = [current_word]
    for _ in range(num_words):
        next_words = model.get(current_word)
        if not next_words:
            break
        next_word = random.choice(next_words)
        output.append(next_word)
        current_word = next_word
    return " ".join(output)
print("\nGenerated Text:")
print(generate_text("natural", 8))
# ♠ spaCy Tokens
doc = nlp(text)
print("\nspacey Tokens:")
print([token.text for token in doc])

```

OUTPUT:

Generated Bigrams:

```
[('natural', 'language'), ('language', 'processing'),
('processing', 'is'), ('is', 'a'), ...]
```

Generated Text:

natural language processing is a field of artificial intelligence

spacey Tokens:

```
['Natural', 'language', 'processing', 'is', 'a', 'field', ...]
```